NIZAMUDDIN ALI AHMED

SENIOR MOBILE ENGINEER

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ANDROID • FLUTTER • ML

Kotlin | Dart | Jetpack Compose
Computer Vision | TensorFlow Lite
Clean Architecture | AI / LLM

PROFILE

Senior Mobile Engineer with 6+ years of professional experience building and modernizing production Android applications, with hands-on Flutter/Dart and applied ML experience. Strong in Kotlin, Java, Jetpack Compose, Clean Architecture, MVVM, modularization, Coroutines/Flow, performance optimization, reusable SDKs, REST integrations, computer vision, TensorFlow Lite, and AI/LLM-backed products. Experienced in technical ownership, iterative algorithm development, production debugging, mentoring, and CI/CD.

CORE EXPERTISE

MOBILE	Android SDK, Flutter, Jetpack Compose, Material 3, Camera, responsive cross-platform UI
LANGUAGES	Kotlin, Dart, Java, SQL
ML / VISION	Pose landmark detection, TensorFlow Lite, exercise recognition, sensor fusion, personalization pipelines
ARCHITECTURE	MVVM, MVI, Clean Architecture, SOLID, modularization, multi-module and feature-first design
DATA & NETWORK	Retrofit, OkHttp, Dio, REST, JSON, GraphQL, Room, SQLite, Firebase, Supabase, Ktor
QUALITY	JUnit, MockK, Flutter testing, debugging, GitHub Actions, CI/CD, Android Profiler, Gradle

EXPERIENCE

VANTAGE CIRCLE

Senior Android Developer | Guwahati, India | Jul 2020 - May 2026

- Led migration of more than 60% of a legacy MVP codebase to MVVM, improving maintainability, separation of concerns, and developer onboarding.
- Migrated 25+ production screens from XML to Jetpack Compose and introduced Material 3 across multiple Android modules.
- Designed feature-based modular architecture for Classifieds and MyVouchers, enabling parallel development and clearer feature ownership.
- **Engineered an adaptive camera-based exercise recognition system**: evolved squat detection from pose-landmark waveform rules to TensorFlow Lite classification for squats, push-ups, and jumping jacks; added posture, lighting, device-orientation, and sensor-based anti-cheat validation.
- Designed a separate personalization feedback pipeline that used completed exercise-session movement data to adapt recognition to individual user movement patterns over time.
- Diagnosed rendering and UI-thread bottlenecks with Android Profiler and implemented Kalman filtering for location/step detection and ML Kit pose-based workout feedback.
- Migrated SQLite persistence to Room and contributed to Android 15 readiness including edge-to-edge UI, photo picker migration, and media-permission updates.
- Built integrations with Retrofit, OkHttp, Apollo GraphQL, Coroutines, Flow, Dagger 2, Firebase, and Android Jetpack; reviewed code and mentored engineers.

GEEKWORKX TECHNOLOGIES

Web and App Developer | Guwahati, India | Jun 2018 - Jun 2020

- Developed and maintained location-based Android applications including Wify Cab and Ind Cab, contributed to a web ERP platform, integrated REST/JSON services, resolved production defects, and mentored junior developers.

KBG SOFTWARE

Android Developer | Guwahati, India | Apr 2018 - Jun 2018

- Maintained Android applications for school clients, implemented features, fixed defects, and integrated REST APIs.

SELECTED ENGINEERING CASE STUDY

ADAPTIVE EXERCISE RECOGNITION

Computer Vision • TensorFlow Lite • Android Sensors • Personalized ML

Built and iteratively hardened a camera-based workout recognition pipeline. The first version used pose landmarks and a waveform-style movement cycle with a ~90% confidence gate. Real-world testing exposed adversarial cases including users exercising while lying down, changing phone orientation, poor-light landmark jitter, and replaying exercise videos.

- **Iteration 1:** Pose landmarks → joint tracking → waveform/cycle detection → confidence threshold → rep count.
- **Anti-cheat:** Added standing-posture validation and Android device-orientation gates after users discovered ways to mimic valid movement.
- **Input quality:** Added lighting validation to bypass recognition when unstable landmarks could create false repetitions.
- **ML evolution:** Replaced hand-written waveform classification with a TensorFlow Lite model trained for squats, push-ups, and jumping jacks.
- **Personalization:** Built a separate training/adaptation loop using completed-session movement data so recognition could progressively adapt to an individual user's movement characteristics.
- **Replay defense:** Used independent Android device sensor/context signals as a lightweight liveness check rather than relying solely on camera output.

Engineering principle: Build → Observe → Break → Understand → Improve → Repeat.

SELECTED PROJECTS

TAPORI AI

Android & Flutter AI Application | 2025-2026

Built and released an LLM-backed conversational product with contextual chat, authentication, credits, payments, Supabase Edge Functions, and production deployment. Migrated the native Android client to Flutter/Dart while preserving Clean Architecture boundaries; used Riverpod, Dio, and GoRouter.

ARCHGUARD

Gradle Plugin & Developer Tool | 2026

Designed an open-source Gradle plugin that converts Android architecture conventions into executable build rules, with a Kotlin/Gradle DSL for required layers, feature-first structures, forbidden folders, validation, and HTML reporting.

SONICBRIDGE

Android TV & Mobile Audio Streaming | 2026

Built Android TV transmitter and mobile receiver apps for low-latency local-network audio streaming using Kotlin, Jetpack Compose, AudioPlaybackCapture, WebSockets, 48 kHz PCM, AudioTrack, and automatic device discovery.

BIOMETRIC SDK

Reusable Android Authentication Library | 2025

Developed a reusable lifecycle-aware Android authentication library around BiometricPrompt with a simplified integration API and consistent success, error, and cancellation handling.

TECHNICAL LEADERSHIP

- Developed reusable mobile SDKs, Gradle tooling, cross-platform Flutter work, ML-backed mobile features, and production applications spanning authentication, AI, real-time networking, and developer experience.
- Created GitHub Actions workflows for Android builds and delivery automation and worked with modular Gradle configurations across Android projects.
- Published technical articles on Android engineering and mentored developers through architecture migrations, Compose adoption, code reviews, debugging, and implementation planning.

EDUCATION

BACHELOR OF ENGINEERING - MECHANICAL ENGINEERING

Royal School of Engineering and Technology | 2013 - 2017

Academic projects: four-legged robot; virtual stress analysis of a bicycle frame.

LINKS

Portfolio: noncoderf.github.io/portfolio.github.io **GitHub:** github.com/NonCoderF **Medium:** medium.com/@sallyinfo365